# NYA-INTEL-SYNTH-001 — Deterministic Intelligence Synthesis

**Vertical slice specification · Question type Q-ASSOC-1 (“source-established relationships around X”)**

| | |
|---|---|
| Document ID | NYA-INTEL-SYNTH-001 |
| Version | **DRAFT v0.5.0** — final drafting pass. Carries the **proposed** founder decision record (§14); nothing is authorised until the founder records the rulings. |
| Supersedes | DRAFT v0.4.0, v0.3.0, v0.2.0, v0.1.0 (all 2026-10-12) |
| Classification | INTERNAL |
| Programme | Nyansapo — Intelligence plane |
| Review basis | Three code-grounded reviews (v0.1.0, v0.2.0, v0.3.0) and a proposed founder decision record on v0.4.0 against `origin/main` `62ead3c54cf984c37d17d9d2b1f4d7b38d5d19ea` (11 Oct 2026); evidence manifests `…-code-evidence.json`, `…-v02-evidence.json`, `…-v03-evidence.json` (the last records three isolated counterexamples). Live data, deployed state and the published PR bytes were not verified by any review. |
| Depends on | NYA-PROVENANCE-EXPLORER-001 PR-0; `applicability_sufficiency.py` evaluation semantics; `governed_verification_service.py`; `KeyManager`; `AuditChain`; ingest prerequisites **IS-P0, IS-P2, IS-P3** (§5.0) |
| Decision effect | `CONTEXT_ONLY`. `decision_effect_allowed = FALSE`, hardwired, not flag-controlled. |
| Author | Prepared with Claude for Omaida Ali, 2026-10-12 |

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# 0. Change log

### 0.1 v0.4.0 → v0.5.0 (final pass — proposed decision record)

| # | v0.4.0 defect | Finding | v0.5.0 disposition |
|---|---|---|---|
| U1 | §10.1 computed the header from latest releases and officer windows **before** the idempotency lookup; `execution_id` included the bundle digest, which does not exist at that point. | A retry after release R2 is admitted cannot recover its original execution by `as_of` alone. | Client-scoped immutable request mapping `intel_request_map (client_id, request_key) → assessment_id` consulted **first**; retrieval by `assessment_id` always available; a new evaluation is a new assessment. `append_in_tx` named explicitly in the atomic path (§10.1, §9.1). |
| U2 | Projection admissibility `parsed + exceptions == artifact_row_count` treated as degree completeness. | 49 parsed at an address + 2 unattributable exceptions passes a threshold of 50 unsafely. | Address degree carries a **lower and upper bound**; `PASS` iff upper ≤ 50, `FAIL` iff lower > 50, else `UNKNOWN`. Exceptions retain identity and content; duplicate-company rows are exceptions; projection is **sealed** (no INSERT/UPDATE/DELETE after `sealed_at`, content digest + parser/normaliser version bound) (§5.0 IS-P2). |
| U3 | `observed_current_through = W_p.end` for every item. | An item on page 1 at `d1` is not observed through the last page's `d2`. | Per-item `observed_through = timestamp of the page the item appeared on`; `W_p` retained as acquisition metadata only; same item with different content across pages → `UNKNOWN(DRIFT)` (§5.0 IS-P2, §5.4). |
| U4 | FD-IS16 tuple presented as appointment identity. | The appointment-list schema makes `appointed_on` and `name_elements` optional and documents no item-level stable id. | Tuple is a **matching key**; canonical full-item digests retained; equal-key/different-digest → `UNKNOWN(IDENTITY_AMBIGUOUS)`; null key components → `UNKNOWN(TUPLE_INCOMPLETE)`; identical digests are duplicates counted once; fixtures added (§5.0 IS-P2, A5b). `rule_status[]` carries `binding_id` and the bound values (§5.5, §6.2). |
| U5 | §13 retired the rule family automatically after two failed re-rulings; IS-P3(b) still appeared as a selectable v1 alternative; stale labels and the register cross-reference. | Retirement is a founder act; (b) is a distinct weaker contract. | Automatic retirement replaced by a mandatory founder disposition (retain blocked / authorise redesign / retire). IS-P3(b) removed from the v1 acceptance contract and kept only in decision history. Labels and cross-references fixed (§5.0, §13.2). |

| # | v0.4.0 defect | Finding | v0.5.0 disposition |
|---|---|---|---|
| U6 | §14 listed recommendations. | The founder has not yet ruled. | §14 is now a **proposed decision record**: ruling text and implementation contract per decision, each marked *proposed — awaiting founder signature*. Nothing in it is treated as adopted by this document. |

### 0.2 v0.3.0 → v0.4.0 (third review)

| # | v0.3.0 defect | Finding (counterexample where executed) | v0.4.0 disposition |
|---|---|---|---|
| T1 | IS-P2 officer degree accepted `total_results == items acquired` as complete. | Duplicate page acquired twice: 100 == 100 passes with 50 distinct items. | Distinct-item coverage under a declared appointment identity; page offsets, parameters and response digests retained; duplicate-page, changing-total and drift detection → `UNKNOWN`; unfiltered history fetched; explicit counting predicate; named per-officer observation window `W_p` (§5.0 IS-P2). |
| T2 | Address membership and degree read from `silver_uk_companies WHERE snapshot_id = R`. | `_handle_company` UPDATE replaces `registered_address` while retaining `snapshot_id`/`source_record_id`; a row labelled R can hold a later streamed address. | Addresses and degree population come **only** from a release-scoped immutable projection built once from the checksum-verified bulk artifact, with parsing-completeness evidence. The evolving Silver table is never the release (§5.0 IS-P2, §5.4). |
| T3 | Ongoing appointment capped at observation time in a half-open interval. | `[1 Jan, 1 Oct)` with `t_R = 1 Oct` excludes the boundary although tenure exceeds 30 days. | `resigned_on` stays an exclusive source endpoint; “observed current through” is a separate **inclusive** censoring boundary; membership at `t_R` defined for both; boundary test cases (§5.4). |

| # | v0.3.0 defect | Finding (counterexample where executed) | v0.4.0 disposition |
|---|---|---|---|
| T4 | “Officer release snapshot” assumed without a defined artefact. | IS-P2 is asynchronous per officer; no single officer release exists. | Replaced by the per-officer observation window `W_p` from IS-P2 acquisition, bound to the appointment facts and degree evidence (§5.0, §5.4). |
| T5 | C14 rejected a retry whose `as_of` predates a release admitted after the original execution. | Retrieval was gated like a new execution. | Required releases pinned at request admission; `(client_id, execution_id)` retrieval precedes the gate; future `as_of` rejected for new executions (§10.1, C14). |
| T6 | §4.2 allowed `UNKNOWN` only with no candidate fact, but §5.4 needed `UNKNOWN` for a present fact missing a date. | Existence and derived-predicate evaluability conflated. | `UNKNOWN` carries a reason: `NO_CANDIDATE` or `FIELD_MISSING`; a fact can be `TRUE` for existence while a derived predicate over it is `UNKNOWN` (§4.2). |
| T7 | Rule 002 has no hypothesis form, so “address evidence missing” had no terminal status and rule 004 could not express its intended hypothesis. | §7.6 “otherwise nothing” is not a status. | Every evaluated `(rule, binding)` gets a terminal status: `EMITTED`, `HYPOTHESIS`, `UNRESOLVED`, `CLOSED`, `SUPPRESSED`. `UNRESOLVED` maps to dependency `UNKNOWN` (§4.3, §5.5, §7). |
| T8 | “Any admissible candidate disproves → FALSE” applied universally. | A non-overlapping historical tenure would negate a separate qualifying tenure for the same pair, or fabricate a conflict. | Aggregation scoped to bound variables (existential): non-matching alternative bindings are ignored; `FALSE` only when the **same** binding is directly disproved; same-binding incompatible values → `FALSE` + rule 005 (§4.2). Mirrors `_eval_paths` keeping alternative paths distinct. |

| # | v0.3.0 defect | Finding (counterexample where executed) | v0.4.0 disposition |
|---|---|---|---|
| T9 | §13.0 “benchmark chain id” and admin purge route. | `audit_chain` has no chain-id column; a different lock id cannot partition it; purging contradicts append-only retention and the zero-grant gate. | Benchmarks run in a **disposable isolated database** with its own audit table; signed acceptance bundle and replay inputs exported before disposal; no purge authority on production tables (§13.0). |
| T10 | `mode` lived outside the signed body. | A receipt could not attest which purpose it served. | `mode` is in the execution header and in the signature domain prefix (§6.2, §11.2). |
| T11 | §11.3 implied all chain callers must supply transactions. | Existing callers own their connections. | Legacy `append` becomes an owning-connection wrapper delegating to a shared serialised helper; `append_in_tx` delegates with the caller’s connection; both take the same lock; all writers re-checked at implementation (§11.3). |
| T12 | Deferred register used “blocked” in place of approval status. | Blocked is an execution state, not an approval state. | Register carries separate **Approval** and **Execution** columns (§0.5). |
| T13 | A7 would have measured an inconsistent interval definition. | Definition-induced misses are not evidence about association quality. | Input-feasibility census (A6a) precedes benchmark assembly; A7 runs only on the corrected temporal rule (§13.2). |
| T14 | FD-IS15(a) read as “attach a batch id”. | A label is not provenance. | (a) binds the **source-response content** to the signed batch (§5.0 IS-P3). |

### 0.3 v0.2.0 → v0.3.0 (follow-up review)

| # | v0.2.0 defect | Finding | v0.3.0 disposition |
|---|---|---|---|
| S1 | Signed body omitted the execution header; §6.1 and §10.1 disagreed on where `as_of` lives. | Replay of a stored bundle must bind everything evaluation consumed. | Execution header ( `question_type` , `subject_key` , `as_of` , rule-set/engine/policy/normalisation digests) is **inside** the canonicalised body (§6.2). |

| # | v0.2.0 defect | Finding | v0.3.0 disposition |
|---|---|---|---|
| S2 | Bundle stored facts but not the decisions made over them. | Normalisation, admissibility, freshness, degree observations and guard inputs were recomputed at replay from live state. | Bundle stores **decisions as well as facts** (§6.4). Replay makes no live DB, clock, investigation or coverage lookup (C8 restated). |
| S3 | `engine_sha256` over a source directory. | Evaluator imports normalisation/policy code from elsewhere; a directory digest does not identify the executable closure. | Replay evaluator depends only on the stored bundle and retained rule definitions. `engine_artifact_sha256` identifies a locked, retrievable build artifact (§6.1). |
| S4 | Below-floor evidence classified `UNKNOWN`, which could spawn a hypothesis; contradicted C11. | Inadmissible evidence is not silence. | New precondition value `INADMISSIBLE_ONLY`; closes the rule; never a hypothesis. Aggregation rule when admissible and inadmissible candidates coexist defined (§4.2). |
| S5 | Failed `DOWNGRADE` guard emitted a hypothesis with no `UNKNOWN` precondition. | A disqualification is not an evidence gap. | `DOWNGRADE_TO_HYPOTHESIS` removed from the schema. All guard failures suppress in v0.3 (§5.2). |
| S6 | Rule 004 treated an upstream rule that emitted nothing as `UNKNOWN`. | A dependency suppressed or disproved upstream would be re-hypothesised downstream. | Upstream status propagates through the DAG: `CLOSED` and `SUPPRESSED` are distinct dependency values that close/suppress the consumer (§5.5). |
| S7 | Rule 002 required a common `observed_at = t` supplied by the request's `as_of`. | Independently acquired current rows do not become contemporaneous by being stamped with one timestamp. | Co-location is evaluated **within a named admitted company release**; `t` is that release's snapshot boundary. Historical `as_of` earlier than the latest admitted release of any required source is **rejected** (§5.4, §10.1). |
| S8 | Appointment interval semantics only covered both-ends-open. | A missing start with a known end was unspecified. | Full date semantics declared: missing start → `UNKNOWN`; missing end → current as of the officer release snapshot (source semantics) (§5.4). |
| S9 | Degree guard relied on "population completeness asserted" via `registry_source_releases`. | Ledger has `row_count` and `PASSED`, but no reconciliation against an upstream population; officer backfill is unpaginated. | New prerequisite **IS-P2** (degree evidence): officer degree from a fully paginated `/officers/{id}/appointments` with `total_results`; address degree from a bulk company release with explicit population and snapshot boundary. Frozen into the bundle. Release admission ≠ completeness (§5.0, §5.2). |
| S10 | `execution_id UNIQUE` globally vs per-client rows. | Second client's row rejected (demonstrated). | `UNIQUE (client_id, execution_id)` (§9.1). |

| # | v0.2.0 defect | Finding | v0.3.0 disposition |
|---|---|---|---|
| S11 | `assertion_id` global PK hashed only `(rule, version, subject_refs, predicate)`. | Collides across repeated assessments; indicators not distinct. | Assessment-scoped composite PK; `assertion_key` binds class, precondition/guard id, temporal scope (§9.1). |
| S12 | POST inserted a row before evaluating/signing. | Runtime has no `UPDATE` grant; a reserved row cannot be completed. | Compute the complete immutable row, then one atomic insert; on conflict, retrieve the authorised existing row (§10.1). |
| S13 | Server-defaulted `as_of` at request time breaks retries. | Each retry gets a new execution id. | Response returns the pinned `as_of`; retries must resend it; unspecified `as_of` is truncated to the minute (§10.1). |
| S14 | Advisory lock on assessment appends only. | Legacy `AuditChain.append` callers do not take it; forks remain possible. | The lock moves **into** `AuditChain.append`; every writer to the chain takes it; concurrency test (§11.3). |
| S15 | `require_client_type_in({...})` with a set; wildcard `'*'` bundles bypass audience. | Helper is variadic; wildcard admin keys would pass. | Variadic call; explicit rejection of wildcard bundles; FD-IS12 (§10). |
| S16 | "Active key" asserted; `KeyManager.load_key()` picks newest PEM by mtime. | No purpose/active-status enforcement. | Purpose-aware selection wrapper required; signing refuses without a DB-active `assessment_snapshot` key (§11.2). |
| S17 | `investigation_ref` joined on synthesis fact ids. | Investigations index entities and fields, not fact ids. | Correlation by entity/field, labelled `correlation: ENTITY_FIELD`, carried in the **envelope** (non-canonical) or null (§12.2). |
| S18 | A4 assumed every IS-P0 support reaches a signed batch. | Stream/backfill writers do not route via `raw_entities → batch_provenance`. | New prerequisite **IS-P3**: officer acquisition provenance, or A4 narrows to what is actually batch-bound (§5.0, §13.1). |
| S19 | Engine reads `ACTIVE` rules; phasing activates only after benchmark. | Draft rules could not be benchmarked. | `BENCHMARK` execution mode with a pinned candidate rule set, isolated from the activation boundary, never served to clients (§13.0). |
| S20 | Deferred register implied authority to postpone. | Deferral is sequencing, not approval. | Register carries an approval state per item (now §0.5). FD-IS2/FD-IS6 marked *recommended, awaiting ruling*, not resolved. |
| S21 | Statement that "every rule suppresses" without completeness. | Rule 005 is not degree-guarded. | Corrected: only degree-guarded rules (001, 002, hence 004) suppress; 005 runs. |

### 0.4 v0.1.0 → v0.2.0 (first review) — retained for the record

R1 sibling evaluator, not AOT generalisation · R2 new assessment canonicaliser · R3 investigations are not dual-controlled overrides; overrides deferred · R4 public verify removed · R5 facts materialised for replay (Silver is upserted in place) · R6 `officer_id` is the appointment self-link; IS-P0 person id · R7 no address occupancy intervals; rule 002 → point-in-time · R8 three-valued guards · R9 only `UNKNOWN` spawns hypotheses · R10 rule DAG · R11 migrations auto-discovered · R12 serialised audit append · R13 execution identity by input hash · R14 benchmark rates with intervals.

### 0.5 Deferred register — approval and execution state

"Blocked" is an execution state and never replaces an approval state. Previously approved items remain approved while blocked.

| Item | Dependency | Approval | Execution |
|---|---|---|---|
| Negative assertions + population coverage model (NYA-INTEL-SYNTH-002) | Predicate/population/time completeness evidence | Awaiting founder decision on sequencing | Not started |
| Immutable signed override objects (NYA-INTEL-SYNTH-003) | Reviewer-authority model; fact-id-level investigation targeting | Awaiting founder decision | Not started |
| Registered-address occupancy intervals (IS-P1) | AD01 parsing or snapshot-succession derivation | Awaiting founder decision | Blocked by dependency |
| Inference drift (NYA-INTEL-SYNTH-004) | First slice landed; schema carries predecessor and digest columns from day one | **Committed** architectural objective; implementation authorisation pending | Sequenced after this slice |
| Public supplied-receipt verifier (NYA-INTEL-SYNTH-005) | Key distribution, revocation semantics, bounded input | Awaiting founder decision | Not started |
| Agent / virtual-office address classifier | No classifier exists in code | Not proposed | Blocked; degree guard only |
| IOI / RRI | Separate commitments; RRI risk-neighbourhood route is 501 | **Committed** (separately); unaffected by this document | Unchanged; never rule preconditions |

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## 1. Purpose and framing

Nyansapo serves governed facts and answers one analytic question deterministically — *who owns or controls X* — through AOT. A second evaluator, `applicability_sufficiency.py`, answers legal-applicability questions with typed `TRUE / FALSE / UNKNOWN / INVALID` outcomes. Consumers who need to know *what the admissible evidence, taken together, establishes* about a subject must assemble that picture themselves.

This slice adds a third evaluator alongside those two:

> **A question-scoped deterministic synthesis capability, implemented alongside AOT and legal applicability, reusing compatible governed infrastructure (identity references, signing primitive, audit chain, governed verification, scope enforcement) while preserving each evaluator's evidence vocabulary and authority boundary.**

Its single question in this slice:

> **Q-ASSOC-1:** *What source-established relationships exist around subject X, and what do they establish?*

Output: a signed **Assessment** containing the facts *and decisions* it was computed from, every assertion with a defined epistemic class, a rule id and version, and the fact ids that support or contradict it. The Assessment says what is established, what conflicts, and — mechanically, from rule preconditions that evaluated `UNKNOWN` — what evidence would change the answer. It says nothing negative in this slice.

Framing that must survive every later document:

- AOT's ownership arithmetic and the applicability evaluator's legal-effect precedence are **not** extended. Synthesis cites their outcomes and never recomputes them.
- The Assessment is **context**. It never carries decision effect.
- **No language model** participates in producing an Assessment.
- Replay is **analytic replay of a stored bundle**, not reproduction of acquisition or identity resolution.

## 2. Scope

### 2.1 In scope

1. One question type, `Q-ASSOC-1`, over one subject.
2. Ingest prerequisites IS-P0 (person identity), IS-P2 (degree evidence), IS-P3 (officer acquisition provenance) (§5.0).
3. Rule schema; four governed rules (§5.4); three-valued guards; identity admissibility floor.
4. The Assessment object with execution header and decision-bearing input bundle (§6); epistemic classes (§4); `UNKNOWN`-only partial fire with DAG status propagation (§5.5, §7).
5. Storage (§9); scoped trusted-service endpoints (§10); canonicalisation, purpose-aware signing, chain-wide serialised audit append (§11); Explorer link-only hooks (§12).
6. Isolated `BENCHMARK` environment, feasibility census and the acceptance gate (§13).

### 2.2 Out of scope — explicitly

- Any other question type. Ownership and control remain AOT's.
- Negative assertions and the coverage model.
- Investigation-derived overrides.
- Public verification.
- Address occupancy intervals; two-hop clusters; ACH; drift logic.
- Prose rendering, PDF, LLM involvement.
- External SDK exposure. Trusted-service only.
- Veredis-side consumption and anything scenario-conditioned.

## 3. Constitutional prohibitions

| # | Prohibition | Enforcement |
|---|---|---|
| C1 | No assertion may carry `decision_effect` other than `NONE`. | `Literal["NONE"]`; test. |
| C2 | Association is never promoted to ownership or control. | Rule loader rejects cross-family predicates. |
| C3 | Intent, purpose or culpability is never inferred. | Closed predicate enum. |
| C4 | Graph proximity is not evidence. | Rule schema requires ≥1 non-topological precondition. |
| C5 | An absent record is not evidence. No negative assertion without a validated coverage record. | No negative predicate exists in v0.3; constructor rejects `*_NOT_ESTABLISHED`. |
| C6 | A Ceres passage is never a fact. | Fact ids validated against fact tables only. |
| C7 | No generative model in the production path. | CI grep gate on `api/services/intel_synthesis/`. |
| C8 | The conclusion body is replayable byte-for-byte from the stored bundle and retained rule definitions, with **no live lookup of any kind** (database, clock, investigations, coverage, policy tables). | Replay test runs with the DB connection and clock stubbed to fail. |
| C9 | Every inference inherits the weakest admissible supporting fact. | Computed in the assertion constructor. |
| C10 | Rules are governed objects: versioned, dual-approved via `governed_verification_service`, immutable once active. | Governed activation gate (§9.2). |
| C11 | A `FALSE`, `INADMISSIBLE_ONLY` or `INVALID` necessary precondition closes a rule; none ever becomes an indicator. | Evaluator semantics (§4.2, §7). |
| C12 | An unknown guard value suppresses; a failed guard suppresses. | Three-valued guards; no downgrade path (§5.2). |
| C13 | A suppressed or closed upstream rule never appears as `UNKNOWN` to a dependent rule. | DAG status propagation (§5.5). |
| C14 | A **new execution** pins the latest admitted release of each required source at admission; its `as_of` is neither earlier than those releases' snapshots nor in the future. Retrieval of an existing assessment is never gated on later releases. | Request gate (§10.1). |

| # | Prohibition | Enforcement |
|---|---|---|
| C15 | A release-scoped address or degree fact is read only from an immutable release projection, never from the evolving Silver table. | Loader reads `ukch_company_release_projection` exclusively for rule 002 and `address_degree`; test asserts no Silver access on that path. |

## 4. Epistemic model

### 4.1 Assertion classes

| Class | Definition | Produced by |
|---|---|---|
| `FACT` | Directly supported by an admissible source record in the bundle. | Fact loader. |
| `DERIVED_FACT` | Deterministic transformation of facts (interval overlap; normalised-address equality within one release). | Derivation functions with stable ids. |
| `INFERENCE` | Rule with all preconditions `TRUE` and all guards `PASS`. | Rule engine. |
| `CONFLICT` | Admissible facts assert incompatible values for the same predicate/scope. | Rule 005. |
| `HYPOTHESIS` | Rule with no closing precondition, ≥ its `hypothesis_min_true` `TRUE`, ≥1 `UNKNOWN`, all guards `PASS`. | Rule engine, §7. |
| `INDICATOR` | `indicator_text` of each `UNKNOWN` precondition of a `HYPOTHESIS`. | Rule engine. |
| `GAP` | A required source family was unavailable or stale at `as_of` (dependency gap, not coverage gap). | Fact loader. |
| `UNKNOWN` | The question cannot be answered for a stated structural reason. | Engine entry gate. |

### 4.2 Precondition evaluation values

| Value | Meaning | Effect |
|---|---|---|
| `TRUE` | Under the current **binding** of the rule's variables, ≥1 admissible candidate satisfies the expression and no admissible candidate under the **same binding** disproves it. | Counts toward full match. |

| Value | Meaning | Effect |
|---|---|---|
| `FALSE` | Under the same binding, an admissible candidate directly disproves the proposition. Same-binding candidates with incompatible values → `FALSE` (fail closed) and the pair is handed to rule 005. | Closes the rule for that binding. |
| `UNKNOWN(reason)` | `NO_CANDIDATE`: no fact in the bundle for the precondition's source families under this binding. `FIELD_MISSING`: a candidate exists and is admissible, but lacks a field the expression needs (e.g. `appointed_on`). Existence may be `TRUE` while a derived predicate over the same fact is `UNKNOWN(FIELD_MISSING)`. | May spawn `HYPOTHESIS` / `INDICATOR`. |
| `INADMISSIBLE_ONLY` | Candidates exist but **every** one is below the admissibility floor. | Closes the rule; `data_warnings` entry naming the floor failure. Never a hypothesis. |
| `INVALID` | Cannot be evaluated (schema mismatch, malformed fact). | Closes the rule; `data_warnings`; engine warning counter. |

Aggregation rules: - Preconditions are **existential over bindings**. A rule is evaluated once per candidate binding of its variables (e.g. per `(p, A, B, tenure_A, tenure_B)`). Alternative bindings that simply do not match are ignored; they never produce `FALSE` for another binding and never produce a conflict. A historical non-overlapping tenure does not negate a separate overlapping tenure for the same officer/company pair. This mirrors `applicability_sufficiency.py::_eval_paths`, which keeps alternative factual paths distinct. - Within one binding, admissible candidates decide; inadmissible candidates are ignored when at least one admissible candidate exists. All candidates are recorded in the bundle with their admissibility reason.

### 4.3 Dependency values (consumer rules only)

| Value | Meaning for the dependent precondition |
|---|---|
| `TRUE` | Upstream emitted its `INFERENCE` / `DERIVED_FACT`. |
| `UNKNOWN` | Upstream terminal status is `HYPOTHESIS` or `UNRESOLVED` (missing evidence, §7). |
| `CLOSED` | Upstream closed (`FALSE`, `INADMISSIBLE_ONLY`, `INVALID`). Dependent rule **closes**. |
| `SUPPRESSED` | Upstream suppressed by a guard. Dependent rule **suppresses**. |

Terminal statuses of a `(rule, binding)` evaluation: `EMITTED(class)`, `HYPOTHESIS`, `UNRESOLVED(reasons)`, `CLOSED(reason)`, `SUPPRESSED(guard)`. Every evaluation has exactly one; "nothing emitted" is never a status on its own.

### 4.4 Admissibility states

`ADMISSIBLE` · `ADMISSIBLE_ADVISORY` · `INADMISSIBLE_IDENTITY` · `INADMISSIBLE_STALE`. (`INADMISSIBLE_OVERRIDDEN` reserved for NYA-INTEL-SYNTH-003.) The state and its reason are **decided at evaluation and stored in the bundle**; replay reads them, it does not recompute them.

### 4.5 Predicate vocabulary (closed)

```
SHARED_OFFICER_ESTABLISHED            (rule 001; INFERENCE)
REGISTERED_ADDRESS_COLOCATED_IN       (rule 002; DERIVED_FACT, within one company
release)
ASSOCIATION_CLUSTER_SUPPORTED         (rule 004; INFERENCE)
RELATIONSHIP_EVIDENCE_CONFLICTED      (rule 005; CONFLICT)
OWNERSHIP_PATH_CITED                  (pass-through of a bounded AOT outcome)
SANCTIONS_LISTING_CITED               (pass-through of sanctions_entity_v1)
```

The AOT adapter inspects the bounded outcome and pinned policy context of the `aot_path_audit` entry; a rejected or insufficient traversal is not cited.

## 5. Rules

### 5.0 Ingest prerequisites

**IS-P0 — person-level officer identity.** Both writers retain raw JSON (`ukch_stream_silver.py::_handle_officer` stores the event envelope; `ukch_officers_charges_backfill.py::_upsert_officer_item` stores the item). Add `silver_uk_officers.ch_person_id` (nullable) extracted from `raw_payload.data.links.officer.appointments` (stream rows) or `raw_payload.links.officer.appointments` (backfill rows), retaining the original link string and its source path. The existing `officer_id` (appointment self-link) is **not** repurposed. Backfill locally where the payload carries the link; a read-only census first classifies rows as *link retained / link absent / malformed / conflicting*, and only the absent class is a candidate for re-fetch. Tests: same person across two companies → one key; two same-name people → two keys; missing link; malformed link; UPDATE path preserves the key. `ch_person_id` is treated as a **source-qualified identifier**, not a proof that every record of one real person shares it.

**IS-P2 — degree evidence.** `registry_source_releases` carries `row_count`, `dlq_count`, `reconciliation_status`; `record_reconciliation` accepts whatever its caller passes and `admit_release` requires only `PASSED` + a timestamp. That is an admission record, not a completeness proof. The officer backfill makes one unpaginated call (`_ch_get`, no pagination loop; `None` on failure coerced to `{}`). And the company stream writer (`ukch_stream_silver.py::_handle_company`) **updates `registered_address` in place while retaining the prior `snapshot_id` and `source_record_id`** (reproduced in isolation), so `WHERE snapshot_id = R` on `silver_uk_companies` does not establish that a row's address came from R. Therefore:

*Officer degree — per-officer paginated acquisition (bounded alternative to acquiring the officer universe).* - Acquire `/officers/{ch_person_id}/appointments` **unfiltered** (no `filter=active`; active-at-`t` is computed from dates), page by page, retaining per page: `start_index`, `items_per_page`, request parameters, response digest, HTTP outcome, `total_results` as reported on that page, ETag if present, timestamp. - **Distinct-item coverage** under a **versioned, acquisition-scoped matching policy** [FD-IS16]. Every raw item is

retained with a canonical full-item digest. Matching key = `(appointed_to.company_number, appointed_on, officer_role, name_elements digest)`; it is a *matching key*, not a proven unique appointment identity (the official appointment-list schema makes `appointed_on` and `name_elements` optional and documents no item-level stable id; the item link is to the company, and the company-officer self-link retained by the existing writer is not assumed to appear here). Rules: identical digests → one item (duplicate); equal key with different digests → `UNKNOWN(IDENTITY_AMBIGUOUS)`; any null key component → `UNKNOWN(TUPLE_INCOMPLETE)`; a page byte-identical to another → `UNKNOWN(ACQUISITION_INCOMPLETE)`; `total_results` differing between pages, any failed page, or distinct items ≠ `total_results` → `UNKNOWN(ACQUISITION_INCOMPLETE)`. A count-only check is explicitly insufficient (counterexample: same 50-item page twice passes `100 == 100`). Stable counts and no duplicate pages are drift checks, not proof of an atomic snapshot. - **Per-item observation**: each item's `observed_through` = the timestamp of the page on which it was acquired. An item on page 1 is **not** observed through the last page's timestamp. If the same item (by digest or key) appears on two pages with different content → `UNKNOWN(DRIFT)` for that officer's degree. - **Acquisition window `W_p`** = `[first page timestamp, last page timestamp]`, retained as metadata only; it never substitutes for per-item `observed_through`. - **Counting predicate**: `officer_degree(p, t)` = number of **distinct company numbers** with an appointment of *p* whose tenure contains `t` under §5.4 membership rules (each item censored at its own `observed_through`), across all roles. Appointments with `UNKNOWN(FIELD_MISSING)` tenure count toward an `uncounted` figure; if `uncounted > 0` the degree is `UNKNOWN(TENURE_UNKNOWN)`. - Limitation stated in the bundle: this is evidence about a Companies House officer key, not proof that all records of one real-world person are unified under it.

*Address membership and degree — release-scoped immutable projection.* - For each admitted bulk company release `R`: verify the retained artifact against `artifact_checksum`, parse it **once** into `ukch_company_release_projection (release_id, company_number, registered_address_norm, normalization_version, company_status, parsed_at)` with `PRIMARY KEY (release_id, company_number)`. Record `population_def`, `snapshot_at`, `artifact_row_count`, `parsed_row_count`, parser version, normaliser version, and an **exceptions table** `ukch_company_release_exceptions (release_id, artifact_row_ref, reason, retained_content_digest, candidate_address_norm NULL)` — a duplicate company number in the artifact is an exception, not a silent overwrite. The projection is **sealed** when `parsed_row_count + exception_count == artifact_row_count`: `sealed_at` and `content_digest` are recorded, and a trigger rejects **INSERT, UPDATE and DELETE** on both tables for that `release_id` thereafter. An unsealed projection is inadmissible. - Rule 002 pair facts and `address_degree` are read **only** from a sealed projection for the same `R` (C15). The evolving Silver table is never the release. - `address_degree(addr_norm, R)` is a **bounded** observation: `lower` = active rows in projection `R` with that `registered_address_norm`; `upper` = `lower` + exceptions for `R` whose `candidate_address_norm` equals `addr_norm` or is null (unattributable). Guard: `PASS` iff `upper ≤ 50`; `FAIL` iff `lower > 50`; otherwise `UNKNOWN(DEGREE_BOUNDS)`. Both bounds, the exception refs and the projection `content_digest` are frozen into the bundle. Parsing reconciliation is not degree completeness.

*Common.* Fix the officer backfill to paginate and to distinguish *empty* from *failed*. Every degree observation (value, method, evidence refs, population or `W_p`, reasons) is frozen into the bundle. Release admission ≠ completeness.

**IS-P3 — officer acquisition provenance.** The stream and backfill writers do not route through `raw_entities → batch_provenance`. The v1 acceptance contract is **(a)**: IS-P0/IS-P2 acquisition is wrapped in a signed batch whose Merkle root **binds the source-response content** (retained page payloads and officer items), not merely a `batch_id` label on the row. When wrapping already-retained historical payloads, the original acquisition

metadata (timestamps, parameters, digests) is preserved; the batch signing timestamp is not a new acquisition timestamp. If (a) is blocked, the dependency is recorded and the slice waits; a weaker `UNBATCHED` guarantee is **not** part of this contract and would require a distinct founder ruling (decision history: FD-IS15, §14).

### 5.1 Rule schema

```yaml
rule_id: NYA-INT-ASSOC-001
version: 1
family: ASSOC
predicate: SHARED_OFFICER_ESTABLISHED
status: DRAFT | ACTIVE | RETIRED
depends_on: []
preconditions:
  - id: P1
    kind: fact | derived | cited | dependency
    source_families: [uk_ch_officers]
    expression: "<closed evaluator expression>"
    admissibility_floor: ADMISSIBLE_ADVISORY
    necessary: true
    indicator_text: "<evidence that would satisfy this precondition>"
guards:
  - id: G1
    kind: degree | class | temporal
    expression: "<closed evaluator expression>"
    on_fail: SUPPRESS            # the only value in v0.3
    on_unknown: SUPPRESS         # the only value in v0.3
emits_on_full_match: INFERENCE | DERIVED_FACT
hypothesis_min_true: <int>
governance:
  verification_subject_table: intel_rules
  author_role: INTEL_RULE_AUTHOR
  reviewer_role: INTEL_RULE_REVIEWER
```

The expression evaluator is a sibling of `applicability_sufficiency.py`'s predicate evaluator: closed, typed, no free text; comparison, boolean, set membership, interval overlap, count; the five precondition values and three guard values. Shares test patterns with `tests/nyansapo/test_applicability_sufficiency_v2_001.py`; extends neither that module nor AOT.

### 5.2 Guards — three-valued, suppress-only

| Guard | Definition | Value |
|---|---|---|
| `officer_degree` | `officer_degree(p, t_R)` per the IS-P2 counting predicate. | `PASS` ≤ 20; `FAIL` > 20; `UNKNOWN` if acquisition incomplete or any tenure unknown. |
| `address_degree` | Bounded `address_degree(addr_norm, R)` from the sealed projection per IS-P2. | `PASS` iff `upper ≤ 50`; `FAIL` iff `lower > 50`; `UNKNOWN` otherwise or if the projection is unsealed. |
| `officer_role_class` | Role ∈ { `secretary`, corporate secretary service}. | `PASS` otherwise; `FAIL` if in set; `UNKNOWN` if role missing. |

| Guard | Definition | Value |
|---|---|---|
| `temporal_overlap_min_days` | Overlap of appointment intervals per §5.4 date semantics. | `PASS` ≥ 30; `FAIL` < 30; `UNKNOWN` if either interval is `UNKNOWN`. |

`FAIL` and `UNKNOWN` both suppress. Guard inputs and outcomes are frozen in the bundle. [FD-IS3.]

### 5.3 Identity admissibility floor

Person: admissible only via `ch_person_id` (IS-P0). Company: `entity_lineage` method in an explicit **allow table** (registration number, LEI) or name method with `match_confidence ≥ 0.90` **and** `registry_status_code = CONFIRMED_REGISTRY_IDENTITY`; methods in the **deny table** (including `alias_canonical_rev`) are inadmissible. The decision and its basis are stored per fact in the bundle. [FD-IS4.]

### 5.4 The four rules

**Date semantics (all rules).** Dates are ISO-8601 at day precision. A tenure has a start `appointed_on` and **one of two distinct end kinds**:

- `resigned_on` present → tenure `[appointed_on, resigned_on)`; the resignation date is an **exclusive** source endpoint.
- `resigned_on` absent → tenure `[appointed_on, observed_through]` where `observed_through` is **that item's own** page timestamp from IS-P2 (not `W_p.end`); this is an **inclusive censoring boundary**, not a resignation. The bundle records `end_kind: RESIGNED | CENSORED` and the `observed_through` value per item.
- `appointed_on` absent → tenure `UNKNOWN(FIELD_MISSING)`; no start is ever invented; the fact's existence remains `TRUE`.

Membership of a day `t` in a tenure: `appointed_on ≤ t` **and** (`end_kind = RESIGNED` ⇒ `t < resigned_on`; `end_kind = CENSORED` ⇒ `t ≤ observed_through`). Overlap of two tenures is the set of days in both; its length is in whole days, computed with the same membership rule at each end. Required test cases: item observation before, equal to and after `t_R`; true resignation exactly on `t_R` (excluded); both tenures censored; one censored one resigned; an appointment whose content changes between pages with unchanged `total_results` (→ `UNKNOWN(DRIFT)`). The counterexample `[1 Jan, 1 Oct)` vs `t_R = 1 Oct` must evaluate `t_R ∈ tenure` under `CENSORED` and `t_R ∉ tenure` under `RESIGNED`.

**Observation contract (rules 002, 004).** Current registered addresses are compared **only within one named, admitted company release** `R` with snapshot boundary `t_R`, read from the release projection (C15). Equality asserts co-location *as of* `t_R`. The request's `as_of` does not reinterpret current rows as historical observations (C14).

**NYA-INT-ASSOC-001 — `SHARED_OFFICER_ESTABLISHED(A, B)`** — `INFERENCE` P1 (necessary): `ch_person_id` *p* has admissible appointment at A with interval `I_A`. P2 (necessary): *p* has admissible appointment at B ≠ A with interval `I_B`. P3 (necessary): derived overlap `I_A ∩ I_B` ≥ 30 days under the membership rule; `UNKNOWN(FIELD_MISSING)` if either tenure is `UNKNOWN`. Evaluated per binding `(p, A, B, I_A, I_B)`; other tenures of *p* at A or B are separate bindings. Guards: `officer_degree`, `officer_role_class`. `hypothesis_min_true: 2`. P1 ∧ P2 `TRUE`, P3 `UNKNOWN` → `HYPOTHESIS`, indicator "appointment dates establishing ≥30-day overlap".

**NYA-INT-ASSOC-002 — `REGISTERED_ADDRESS_COLOCATED_IN(A, B, R)`** — `DERIVED_FACT` P1 (necessary): A's `registered_address_norm` in projection `R`, admissible. P2 (necessary): B's `registered_address_norm` in the same projection `R`, equal to A's. Guard: `address_degree(addr_norm, R)`. No hypothesis form. If A or B is absent from projection `R` → terminal status `UNRESOLVED(NO_CANDIDATE)`, which rule 004 sees as `UNKNOWN`.

**NYA-INT-ASSOC-004 — `ASSOCIATION_CLUSTER_SUPPORTED(A, B)`** — `INFERENCE` `depends_on: [001, 002]`. P1 (necessary, dependency): rule 001 for (A,B). P2 (necessary, dependency): rule 002 for (A,B) in release `R` **and** `t_R ∈ I_A ∩ I_B` under the §5.4 membership rule (a censored tenure includes `t_R` when `t_R ≤` that item's `observed_through`). `hypothesis_min_true: 1`. One dependency `TRUE`, the other `UNKNOWN` → `HYPOTHESIS` naming the missing one. Either `CLOSED` → closed; either `SUPPRESSED` → suppressed (C13). Never consults ownership (C2).

**NYA-INT-ASSOC-005 — `RELATIONSHIP_EVIDENCE_CONFLICTED`** — `CONFLICT` P1 (necessary): two admissible facts in the bundle assert incompatible values for the same (subject, predicate, release/snapshot), or a precondition elsewhere evaluated `FALSE` with both satisfying and disproving admissible candidates (§4.2). Emits `CONFLICT` citing both facts. Not degree-guarded. Does not resolve.

### 5.5 Dependency DAG and status propagation

Loader builds the DAG from `depends_on`, rejects cycles, evaluates topologically (rule id tiebreak). Each `(rule, binding)` terminal status — `EMITTED(class)`, `HYPOTHESIS`, `UNRESOLVED(reasons)`, `CLOSED(reason)`, `SUPPRESSED(guard)` — is recorded in `rule_status[]` as `{rule_id, version, binding_id, binding: {var: value…}, status, reason}` where `binding_id = sha256(rule_id, version, canonical binding values)`, and mapped to the dependency values of §4.3 for consumers. Lineage closure: a consumer's supporting-fact set is the union of its dependencies' supporting sets.

## 6. The Assessment object

### 6.1 Identity (outside the signed body)

```
assessment_id              asm-YYYY-MMDD-NNNNNN
client_id                  owner; object-level access key
execution_id               sha256(canonical execution header ‖ input_bundle_sha256)
created_at, request_id, caller_scan_id, external_ref
audit_chain_seq, signing_key_id, signature, canonical_sha256
investigation_correlations[]   {assertion_key, investigation_id, correlation:
ENTITY_FIELD}   # non-canonical, §12.2
supersedes_assessment_id   NULL in v0.3
```

### 6.2 Signed conclusion body (canonicalised)

```
canonicalization_version   "assessment-v1"
execution_header
  mode                     SERVE | BENCHMARK
  question_type            Q-ASSOC-1
  subject_key              nya:entity:v1:<hex> | unresolved:{normalised request,
identity_state}
  as_of                    ISO-8601 UTC, minute precision
  rule_set_version, rule_set_sha256
  engine_artifact_sha256   digest of the locked build artifact (wheel + lockfile), not
a directory
  normalization_version    address/name normaliser version
  admissibility_policy_version
  required_releases[]      {source_family, release_id, snapshot_at, artifact_checksum,
projection_admissible}   # pinned at admission
  officer_acquisitions[]   {ch_person_id, W_p.start, W_p.end, matching_policy_version,
acquisition_status}
subject                    canonical_id, golden_id, entity_type, identifiers[],
identity_state, admissibility_state, admissibility_basis
input_bundle               §6.4
derived[]                  DERIVED_FACT entries with derivation id and input fact keys
rule_status[]              {rule_id, version, binding_id, binding{}, status, reason}
# §5.5, one row per evaluated binding
assertions[]               assertion_key, class, predicate, subject_refs[], rule_id,
rule_version,
                           preconditions[{id, value, satisfied_by: fact_key|null,
indicator_text}],
                           guards[{id, value, observed_value, evidence_ref}],
                           supporting_fact_keys[], contradicting_fact_keys[],
                           temporal_scope, admissibility_state, decision_effect "NONE"
gaps[]                     source_family, reason, freshness_state
counts                     facts_by_admissibility{}, assertions_by_class{},
rules_closed{}, rules_suppressed{}
```

### 6.3 Replay contract

Replay = re-run the retained rule definitions (by `rule_set_sha256`) over the stored `input_bundle` and `execution_header`, with **no** database, clock, investigation, coverage or policy-table access (C8). The replay evaluator is a pure function of (bundle, rules). Any decision that would otherwise need live state is stored in the bundle (§6.4). The digest `engine_artifact_sha256` must resolve to a retrievable artifact in the release store.

### 6.4 Input bundle — facts and decisions

Per fact: source table, row key, full typed field set consumed by any precondition or guard, raw `observed`/release reference, `ch_person_id` and original link (officers), lineage ref and `match_method` string, **and the decisions**: admissibility state + reason, freshness state at `as_of` (output of `evaluate_source_freshness` with injected time), normalised values + `normalization_version`. Per guard: observed value, method, evidence refs (paginated appointments summary; company release fields), outcome. Per release: id, snapshot, checksum, population definition.

`input_bundle_sha256` is computed over the canonicalised bundle. Loader scope: one hop — companies sharing a `ch_person_id` or a normalised registered address (within the named release) with the subject.

## 7. Partial-fire mechanics

1. For each rule in DAG order, evaluate preconditions (§4.2/§4.3) and guards (§5.2).
2. Any necessary precondition `FALSE`, `INADMISSIBLE_ONLY`, `INVALID` or `CLOSED` → rule `CLOSED(reason)`; nothing emitted (C11, C13).
3. Any guard `FAIL`/`UNKNOWN`, or any dependency `SUPPRESSED` → rule `SUPPRESSED(guard)`; nothing emitted (C12, C13).
4. All preconditions `TRUE`, all guards `PASS` → emit declared class.
5. No closing value, ≥ `hypothesis_min_true` `TRUE`, ≥1 `UNKNOWN`, all guards `PASS` → emit one `HYPOTHESIS` with the full precondition table and one `INDICATOR` per `UNKNOWN` precondition.
6. Otherwise → `UNRESOLVED(reasons)`: the `UNKNOWN` reasons of the unsatisfied preconditions. Nothing is emitted, but the status is recorded and propagates as dependency `UNKNOWN`.

Every `(rule, binding)` evaluation ends in exactly one terminal status. A hypothesis arises only from silence in the evidence, after every guard has passed; never from contradiction, inadmissibility or disqualification.

## 8. Coverage (dependency, not built)

Negative assertions require coverage evidence that does not exist.
`api/models/coverage.py::DimensionCoverage` is the correct base and will be extended in NYA-INTEL-SYNTH-002. v0.3 emits no negatives (C5). The IS-P2 completeness notions are used only to **suppress**.

## 9. Data model

### 9.1 Tables (additive migration, auto-discovered)

```
intel_rules
  rule_id, version, family, predicate, status, definition_jsonb, definition_sha256,
  verification_request_id, activated_at, retired_at
  PK (rule_id, version); partial unique (rule_id) WHERE status='ACTIVE'

intel_rule_sets
  rule_set_version PK, rule_versions_jsonb, rule_set_sha256, activated_at, mode (ACTIVE
| BENCHMARK)

intel_assessments
  assessment_id PK, client_id, execution_id, mode, question_type, subject_key, as_of,
  rule_set_version, rule_set_sha256, engine_artifact_sha256, input_bundle_sha256,
  body_jsonb, canonical_sha256, signature, signing_key_id, audit_chain_seq,
  created_at, request_id, caller_scan_id, external_ref,
investigation_correlations_jsonb,
  supersedes_assessment_id NULL
  UNIQUE (client_id, execution_id)

intel_request_map
  client_id, request_key, assessment_id FK, created_at
  PK (client_id, request_key)            -- immutable; INSERT only

intel_assertions
  assessment_id FK, assertion_ordinal, assertion_key, class, predicate, rule_id,
rule_version,
  precondition_or_guard_id NULL, temporal_scope_jsonb, admissibility_state,
  subject_refs_jsonb, supporting_fact_keys_jsonb, contradicting_fact_keys_jsonb
  PK (assessment_id, assertion_ordinal); INDEX (assertion_key)
```

`assertion_key = sha256(rule_id, version, class, predicate, subject_refs, precondition_or_guard_id, temporal_scope)`; stable across assessments for correlation, unique only within one.

### 9.2 Grants and governance

Runtime role: `SELECT` all; `INSERT` on `intel_assessments`, `intel_assertions`, `intel_request_map`; no `UPDATE`/`DELETE`. Rule rows written only by the governed activation route: `governed_verification_service` (`create_verification_request` → two typed-role `submit_review`s under `INTEL_RULE_AUTHOR` / `INTEL_RULE_REVIEWER`), then a stage-advancement-style gate adapted to `intel_rules.status`. “Dual-approved” = two recorded typed-role approvals, not two cryptographic signatures. [FD-IS8.]

### 9.3 Migration

Additive. Privilege manifest, zero-grant gate, role tests. IS-P0/IS-P2 migrations on `silver_uk_officers` and a new `ukch_officer_appointments_evidence` table are separate and earlier.

## 10. API

Trusted-service surface. Scope **`INTEL_READ`** added to the closed catalogue; `require_scope("INTEL_READ")` **and** `require_client_type_in("FIRST_PARTY_APPLICATION")` (variadic) **and** an explicit check rejecting wildcard `'*'` bundles (the helper bypasses audience for them, `auth.py:650–681`). [FD-IS12.] Object-level: readable only by the owning `client_id`. Route prefix `/v1/intel/assess/*` (no collision at inspected revision; re-check at wiring).

### 10.1 `POST /v1/intel/assess`

Request: `{question_type, subject: {golden_id | canonical_id | company_number}, as_of?, caller_scan_id?, external_ref?}`.

Admission, in order: 1. **Retry lookup first.** `request_key` = the client-supplied `Idempotency-Key` header, else `sha256(canonical request body)`. If `intel_request_map (client_id, request_key)` exists, return that `assessment_id`'s stored assessment unchanged, with **no** release gate, no repinning and no re-evaluation (retrieval never depends on later releases, rules or facts). `GET /v1/intel/assess/{assessment_id}` is always available as the explicit original-assessment path. 2. Otherwise resolve the subject; pin `required_releases[]` to the latest admitted, **sealed** projection of each required source at this moment; obtain IS-P2 officer acquisitions; apply the gate: `as_of` (default request time truncated to the minute; returned in the response) must satisfy `max(required_releases.snapshot_at) ≤ as_of ≤ now` → else 422 (C14). A future `as_of` is never accepted as evidence that observations extend into the future. 3. Load facts and decisions; build the bundle; compute `input_bundle_sha256` and then `execution_id`; evaluate; canonicalise; sign. 4. In one transaction: insert the complete `intel_assessments` row (on `UNIQUE (client_id, execution_id)` conflict → use the existing authorised row), insert `intel_request_map`, call **`AuditChain.append_in_tx(conn, event)`** (the owning-connection wrapper cannot provide this atomicity), commit.

A new evaluation against current inputs is always a new assessment. Tests: retry after a new release is admitted; after a rule-set activation; after a fact changes — each returns the original.

No row is ever reserved and completed later. Returns the Assessment in `StandardEnvelope`; every inadmissible fact, closed rule and suppressed guard appears in `data_warnings` with `client_action`.

**10.2 `GET /v1/intel/assess/{assessment_id}` — stored body, byte-identical; same access boundary.**

**10.3 `GET /v1/intel/assess/{assessment_id}/explain/{assertion_key}` — rule definition at the cited version; precondition/guard tables; supporting facts from the stored bundle with lineage via `provenance_explorer_service.get_raw_chain`.**

**10.4 `GET /v1/intel/assess/{assessment_id}/verify` — scoped. Recomputes `canonical_sha256`, verifies signature under the recorded key, checks the `audit_chain` event at `audit_chain_seq` carries this digest. Cryptographic validity reported separately from factual correctness.**

**10.5 Errors — 401/403/422/429/503. Unresolved subject or empty fact set → 200 with `UNKNOWN`/empty assessment.**

## 11. Canonicalisation, signing, audit

### 11.1 `canonicalize_assessment_v1`

Explicit field schema for §6.2; sorted keys; nulls pruned; arrays ordered by deterministic keys; numbers as decimal strings; dates ISO-8601 UTC at declared precision. Tests: per-field tamper changes the digest; two distinct assessments never collide; stored body re-canonicalises to its stored digest; the header is covered.

### 11.2 Signing

`KeyManager.sign/verify` as-is for the primitive. `load_key()` selects by filesystem mtime and ignores purpose and DB status, so a **purpose-aware selection wrapper** is required: it resolves the DB-active `signing_keys` row with `purpose='assessment_snapshot'`, loads exactly that key, and refuses (503, no unsigned output) if none is active. Domain separation prefix `NYA-ASSESSMENT-V1:<mode>\0` where `<mode>` is `SERVE` or `BENCHMARK`, so a receipt cryptographically attests its purpose in addition to `mode` being inside the signed header. Rotation policy: new key activated → old key retained for verify only. [FD-IS11.]

### 11.3 Audit chain — chain-wide serialisation

Advisory locks coordinate only callers that take them, and `audit_chain` is one unpartitioned table. Therefore a single serialised helper `_append_locked(conn, event)` takes `pg_advisory_xact_lock(<one constant chain lock id>)` on the supplied connection, reads the latest hash, inserts the successor. The existing `AuditChain.append(event)` becomes an **owning-connection wrapper**: it opens its own connection/transaction and delegates to the helper, so existing callers need no change. The new `AuditChain.append_in_tx(conn, event)` delegates with the caller's connection, which is what assessment insertion uses so the row and its `ASSESSMENT_SIGNED` event commit atomically. Any writer found at implementation time that inserts into `audit_chain` directly (the focused search found the one in `provenance/chain.py`; re-check all) is routed through the helper. Concurrency test: N parallel writers mixing both entry points; chain verifies; no forked predecessor. [FD-IS13.]

## 12. Provenance Explorer integration (link only)

### 12.1 Read surfaces — ASSESSMENT search tab; assessment page; ASSESSMENTS lifeline lane; synthesis stage on the stage chain. Admin-authenticated.

### 12.2 Investigations — correlation, not join

Investigations index entities and fields, not synthesis fact keys. v0.3 correlates a `CONFLICT` assertion to open/concluded investigations by (entity ref, field name) and stores the result in the **envelope** as `investigation_correlations[] {assertion_key, investigation_id, correlation: ENTITY_FIELD}`. It is non-canonical, not signed, and may be recomputed on read. The engine never files investigations.

### 12.3 Overrides — deferred to NYA-INTEL-SYNTH-003. Investigation outcomes do not alter admissibility in v0.3.

## 13. Benchmark mode and acceptance gate

### 13.0 `BENCHMARK` mode

The engine normally reads the `ACTIVE` rule set. For acceptance, a pinned **candidate** rule set (digest recorded) is evaluated in a **disposable isolated database** provisioned for the benchmark run: its own `intel_*` tables, its own `audit_chain` table (the production `audit_chain` has no partition column and cannot host a second chain), and a `BENCHMARK`-purpose signing key. Before the environment is disposed, the signed acceptance manifest, every benchmark assessment body, the input bundles and the candidate rule definitions are exported to the release store as the acceptance bundle. Production `intel_assessments` and `audit_chain` remain append-only; **no purge route exists**. §10 routes are hardwired to `mode = SERVE` and never see benchmark rows because they are in a different database. Activation (DRAFT → ACTIVE) remains a separate governed act after the benchmark.

### 13.1 Correctness

- **A1 Replay.** 100 stored assessments regenerate a byte-identical conclusion body with DB and clock stubbed to fail.
- **A1b Retry.** The same request replayed after a new release admission, a rule-set activation and a fact change returns the original assessment via `intel_request_map`.
- **A2 Constitution.** C1–C15 tests pass; C7 grep gate present.
- **A3 Canonicaliser.** Tamper, collision, header-coverage tests pass.
- **A4 Explain and provenance.** Every `INFERENCE`/`HYPOTHESIS` in a 500-assessment sample resolves to a complete explain payload, and every supporting fact leads to a signed batch whose Merkle root binds the retained source content (IS-P3).
- **A5 IS-P0 fixtures** (§5.0) pass, including missing/malformed link and UPDATE path.
- **A5b IS-P2 fixtures**: duplicate page → `UNKNOWN(ACQUISITION_INCOMPLETE)`; `total_results` changing between pages → `UNKNOWN`; failed page → `UNKNOWN`; distinct-item shortfall → `UNKNOWN`; equal matching key with different item digests → `UNKNOWN(IDENTITY_AMBIGUOUS)`; missing `name_elements` or `appointed_on` in the key → `UNKNOWN(TUPLE_INCOMPLETE)`; equal key and digest repeated → counted once; item content changing between pages → `UNKNOWN(DRIFT)`; any tenure with missing start

→ `UNKNOWN(TENURE_UNKNOWN)`; unsealed projection → inadmissible; exceptions at or unattributable to an address raise `upper` and can turn `PASS` into `UNKNOWN`; write to a sealed projection rejected; a Silver row whose address differs from projection `R` for the same company is **not** read by rule 002 (C15 test).

### 13.2 Analytic quality — the number that decides

Frozen benchmark: selection script, seed, labels, identity evidence, degree evidence and input bundles committed; candidate rule set digest recorded.

- **A6a Input-feasibility census** (precedes assembly). Over the latest admissible projection `R` and a random sample of 10,000 officers with complete IS-P2 acquisition, count officer/company pairs for which the corrected temporal rule can be evaluated at all (`t_R` inside a ≥30-day overlap with both tenures known). Report the count and the distribution of `UNRESOLVED` reasons. If the count is below 400, A7's 200 positives cannot be drawn without bias and the founder is asked whether to widen `R` selection before benchmarking.
- **A6 Negative control.** 200 pairs selected as unrelated + 50 hard negatives (same formation-agent address with degree > 50; same nominee director with degree > 20). Report counts receiving `ASSOCIATION_CLUSTER_SUPPORTED` with 95% Wilson intervals. Gate: ≤ 2/200 on the plain set; 0/50 on the hard set, each showing a guard suppression.
- **A7 Positive control.** 200 pairs hand-verified to share a `ch_person_id` with ≥30-day overlap under the §5.4 membership rule and to be co-located in projection `R` with `t_R` inside the overlap. Runs only on the corrected temporal definition. Gate: ≥ 180 hits. Report separately: misses by guard suppression (each reviewed), by identity floor, by `INADMISSIBLE_ONLY`, and genuine misses. Suppressions are not hits.
- **A8 Predicate-level accuracy** per rule (001, 002) before compound 004.
- **A9 Guard effect** over a 10,000-company random sample: guards off vs on; `UNKNOWN` suppressions reported separately. No target; informs FD-IS3 v2.
- **A10 Reviewer agreement** on 50 random `HYPOTHESIS` assertions: ≥ 80% agree the indicator names the right missing evidence.

### 13.3 Operational

- **A11** p95 `POST` ≤ 2.0 s on QA for ≤ 50 one-hop neighbours (excluding IS-P2 acquisition, which is asynchronous).
- **A12** Zero-grant gate passes for `intel_*`.
- **A13** QA/UAT digests match at the slice commit.
- **A14 Signing readiness receipt**: DB-active `assessment_snapshot` key, one assessment signed and verified via §10.4, event present in `audit_chain`; recorded in the acceptance manifest.
- **A15 Chain concurrency** test passes with both legacy and synthesis writers.

If A6 or A7 fail: thresholds re-ruled under governed rule versions, benchmark re-runs. After two re-rulings still failing, the outcome goes to the founder for an explicit disposition — **retain blocked**, **authorise redesign**, or **retire** — recorded in the decision record. Nothing retires automatically; a failed benchmark does not alter parameters or withdraw an approved capability on its own.

## 14. Proposed founder decision record

Each row is a **proposed ruling** drafted for the founder's adoption. None is adopted by this document; nothing below authorises implementation, activation, key provisioning or deployment until the founder records it. Status of every row: *proposed — awaiting founder signature*.

| ID | Proposed ruling | Implementation contract |
|---|---|---|
| FD-IS1 | **Approve Q-ASSOC-1 and all three ingest prerequisites.** | IS-P0 (source-qualified officer key), IS-P2 (degree evidence + sealed release projection), IS-P3 (content-bound signed acquisition) are mandatory before IS-1. IS-0 may proceed once decisions are recorded. Associations remain context only. |
| FD-IS2 | **Approve a sibling evaluator.** | Adopt the applicability evaluator's typed evaluation patterns with the synthesis-specific values and dependency statuses defined here. Do not generalise AOT's ownership arithmetic or change the legal applicability evaluator. Reuse signing, audit, identity and governance infrastructure only through explicit adapters. |
| FD-IS3 | **Approve 20 companies / 50 active companies per address / 30 days as initial governed benchmark parameters.** | Provisional analytic parameters, not demonstrated accuracy and not a nominee/agent classifier. Unknown or failed guards suppress. Any change is a new governed rule version with revalidation; a failed benchmark neither alters parameters nor retires an approved capability on its own. |
| FD-IS4 | **Approve the strict identity floor with explicit allow/deny method tables.** | Officers via `ch_person_id` only. Company resolver methods mapped explicitly; registration numbers carry their jurisdiction/source context; name-based evidence needs both the confidence floor and `CONFIRMED_REGISTRY_IDENTITY`. Deny `alias_canonical_rev`; **unknown methods fail closed**. Method, policy version and decision frozen in the bundle. |

| ID | Proposed ruling | Implementation contract |
|---|---|---|
| FD-IS5 | **Approve per-client rows, `UNIQUE (client_id, execution_id)`, and request-key retry mapping.** | Execution identity = header + bundle digest; caller references never mutate signed results. A retry locates its original execution through `intel_request_map` or by `assessment_id` **before** any repinning; resending `as_of` alone is not sufficient. A new evaluation against current inputs is a new assessment. |
| FD-IS6 | **Approve scoped verification only.** | Same audience, scope and ownership checks as retrieval; verify stored digest, signature and matching audit event; report cryptographic integrity separately from factual correctness. Public receipt verification remains a separate, untaken decision. |
| FD-IS7 | **Approve `/v1/intel/assess` and child routes.** | Re-check collisions at the implementation revision; keep catalogue/envelope conventions. SDK methods inherit the trusted-service boundary and do not authorise external exposure. |
| FD-IS8 | **Approve two distinct, role-authorised human approvals through existing governance machinery.** | Typed roles `INTEL_RULE_AUTHOR` / `INTEL_RULE_REVIEWER`, both slots filled, existing proposer/self-review exclusions enforced — the author role names an approval role and never permits the proposer to self-approve. Generic untyped activation requests rejected. Approvals bound to the exact definition digest; active definitions immutable. Two approvals ≠ two cryptographic signatures. |
| FD-IS9 | **Approve sequencing coverage and negative assertions after this slice.** | No negative predicates now. Coverage and negatives remain tracked work, not cancelled; prior approval status preserved and execution sequencing recorded separately (§0.5). IS-P2 completeness evidence must still prevent guards passing on undercounts. |
| FD-IS10 | **Approve `REGISTERED_ADDRESS_COLOCATED_IN(A,B,R)` as a release-scoped derived fact.** | Pair equality and bounded address degree read only the sealed projection for the same `R`. Proves registered-address equality in that release — not physical occupancy, operational association or day-current address. Parsing exceptions raise the degree upper bound and never silently lower it. |

| ID | Proposed ruling | Implementation contract |
|---|---|---|
| FD-IS11 | **Approve fail-closed, purpose-aware signing.** | New strict canonicaliser; domain separation with mode; exact DB-active `assessment_snapshot` key selection. No eligible key → 503, never an unsigned assessment. Historical verification material retained. A14 remains the readiness gate; this ruling does not assert a key is deployed. |
| FD-IS12 | **Exclude wildcard `*` bundles from synthesis routes.** | Explicitly scoped first-party credentials plus object-level ownership; explicit wildcard rejection on this surface only; global wildcard semantics and the Explorer's admin surface unchanged. |
| FD-IS13 | **Approve the shared chain-wide advisory-lock helper.** | Legacy `append` owns its connection and delegates; `append_in_tx` delegates with the caller's transaction; one lock; all direct writers participate; assessment row and audit event commit atomically; legacy signatures preserved. A failing concurrency test blocks release until fixed; single-writer is a separate architectural decision. |
| FD-IS14 | **Approve release-bound observations and rejection of unsupported historical/future executions.** | New executions pin their admitted sealed release set; stored retrieval/replay is unaffected. Resignation exclusive, censoring inclusive **at each item's own observed-through bound**; no invented dates; `as_of` cannot extend observed evidence. |
| FD-IS15 | **Choose (a): signed batches binding source-response content.** | Page responses and officer items bound by content hashes/Merkle evidence; original acquisition metadata preserved when wrapping retained payloads. `UNBATCHED` is not a fallback; if (a) is blocked the dependency is recorded and any weaker guarantee needs its own ruling. |
| FD-IS16 | **Approve a versioned, acquisition-scoped distinct-item policy; the tuple is a matching key, not identity.** | Prefer a stable appointment id only where present and verified. Retain canonical full-item digests and page/offset evidence; equal-key/different-item → ambiguous → `UNKNOWN`; null key components → `UNKNOWN`; identical digests deduplicated; every raw item retained. Never deduplicate distinct appointments merely because the key agrees. |

## 15. Phasing

| Phase | Content | Exit |
|---|---|---|
| IS-P0 | `ch_person_id` extraction + census + backfill; identity fixtures. | A5. |
| IS-P2 | Paginated per-officer acquisition with distinct-item coverage and `W_p`; `ukch_company_release_projection` built once per release from the verified artifact; backfill pagination fix. | A5b. |
| IS-P3 | Officer acquisition provenance per FD-IS15. | A4 definition fixed. |
| IS-0 | Migration; `intel_rules`, loader, DAG, sibling evaluator; four rules DRAFT; governed-verification wiring. | Cycle/cross-family rejection tests. |
| IS-1 | Fact + decision loader at `as_of`; bundle + header + hash; admissibility; guards; AOT adapter. | C9, C12, C13 enforced. |
| IS-2 | Engine; assessment construction; `canonicalize_assessment_v1`; purpose-aware signing; chain-wide lock; atomic insert. | A1, A2, A3, A15. |
| IS-3 | Routes; `INTEL_READ` + audience + wildcard rejection + object-level; SDK `assess()`; parity gate. | Contract-complete per catalogue generator. |
| IS-4 | Explorer tab, lane, stage, explain view, correlations. | A4. |
| IS-5 | Isolated benchmark environment; A6a census; A6–A10; A14; exported acceptance bundle; founder ruling keep / re-rule / retire. | Ruling recorded. |

No phase begins before the founder has recorded FD-IS1–IS16. IS-0 may begin before IS-P0/P2/P3 complete; IS-1 may not.

## 16. Glossary

| Term | Meaning |
|---|---|
| Assessment | Signed output of one question over one subject, carrying its execution header and decision-bearing input bundle. |
| Execution header | Canonical block binding question, subject, `as_of`, rule-set, engine-artifact, policy and normalisation digests, and required releases. |

| Term | Meaning |
|---|---|
| Input bundle | Facts **and the decisions made over them** (admissibility, freshness, normalisation, degree, guard outcomes), materialised at evaluation. |
| Analytic replay | Re-evaluation of retained rules over the stored bundle with no live lookup; not reproduction of acquisition or resolution. |
| `TRUE / FALSE / UNKNOWN(reason) / INADMISSIBLE_ONLY / INVALID` | Precondition values per binding; only `UNKNOWN` (`NO_CANDIDATE` or `FIELD_MISSING`) can spawn a hypothesis. |
| `EMITTED / HYPOTHESIS / UNRESOLVED / CLOSED / SUPPRESSED` | Terminal statuses of a `(rule, binding)` evaluation. |
| `W_p` | Per-officer acquisition window (metadata only). Each item is censored at its own page timestamp `observed_through`, never at `W_p.end`. |
| Matching key | Acquisition-scoped tuple used to detect duplicates in IS-P2; not a proven appointment identity. |
| Sealed projection | Release projection with `sealed_at` and `content_digest`; INSERT/UPDATE/DELETE rejected thereafter. |
| `intel_request_map` | Immutable client-scoped mapping from request key to assessment id; consulted before any repinning. |
| `RESIGNED` / `CENSORED` | Tenure end kinds: exclusive source endpoint vs inclusive observation boundary. |
| Release projection | `ukch_company_release_projection`: immutable, once-parsed, checksum-verified view of one bulk release; the only source for rule 002 and address degree. |
| `TRUE / UNKNOWN / CLOSED / SUPPRESSED` | Dependency values seen by consumer rules. |
| `PASS / FAIL / UNKNOWN` | Guard values; `FAIL` and `UNKNOWN` suppress. |
| `ch_person_id` | Source-qualified person-level Companies House officer identifier (IS-P0). |
| Degree evidence | Distinct-item-complete paginated appointments (officer) or admissible release projection (address) (IS-P2). |
| Release `R`, `t_R` | A named admitted company release and its snapshot boundary; the only context in which addresses are compared. |
| `BENCHMARK` mode | Evaluation of a pinned candidate rule set in a disposable isolated database with its own audit table; exported before disposal; bound into the signed header and signature domain. |
| C1–C15 · FD-IS1–16 · A1–A15 (+A1b, A5b, A6a) | Prohibitions · proposed founder rulings · acceptance criteria. |

*v0.5.0 generated 2026-10-12. DRAFT — proposed decision record attached; not authorised for implementation until the founder records FD-IS1–16.*